Niklas Arppe

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niklasarppe

Research interests

Various flavors of the Langlands program (local, global, geometric, categorical), Shimura varieties, Hecke stacks, shtukas, p-adic Hodge theory, motivic homotopy theory

Education

2025–2027 **MSc Mathematics**, *University of Bonn*
Grade: 1.1 (Best possible grade: 1.0)

2023–2025 **BSc Mathematics**, *University of Helsinki*
Grade: 5.0 (Best possible grade: 5.0)

Selected talks

Expository talks

01/2027 **Moduli of local shtukas seminar**, *University of Bonn*
Title: Local Shimura varieties

01/2027 **Kan seminar**, *University of Bonn*
Title: On the cohomology and K-theory of general linear groups over a finite field

12/2026 **Topological cyclic homology seminar**, *University of Bonn*
Title: The Nygaard filtration and syntomic cohomology

10/2026 **Topological cyclic homology seminar**, *University of Bonn*
Title: Cyclotomic spectra

10/2026 **Moduli of local shtukas seminar**, *University of Bonn*
Title: Perfectoid spaces

07/2026 **Simplicial homotopy theory seminar**, *University of Bonn*
Title: The Dold–Kan correspondence

07/2026 **Algebraic K-theory seminar**, *University of Bonn*
Title: The $+$=S theorem and dévissage arguments

06/2026 **Prismatic F-gauges seminar**, *University of Bonn*
Title: Filtered prismatization and gauges

02/2026 **Higher category theory seminar**, *University of Bonn*
Title: Spectra and stable ∞-categories

01/2026 **Absolute prismatic cohomology seminar**, *University of Bonn*
Title: Relative prismatic cohomology

10/2025 **Absolute prismatic cohomology seminar**, *University of Bonn*
Title: Recollections on classical prismatic cohomology

07/2025 **Derived algebraic geometry seminar**, *Aalto University*
Title: Basics on non-derived stacks

Selected awards

06/2022 **Technology Industries of Finland Centennial Foundation's stipend**

06/2022 **The Håkan Simberg Foundation's stipend**

Conference and workshop attendance

11/2026 **Moduli spaces and beyond**, *University of Bonn*

10/2026 **Workshop on Beilinson–Drinfeld Quantization**, *Technical University Darmstadt*

10/2026 **Kleine AG: The isomorphism between the Drinfeld and Lubin–Tate spaces at infinite level**, *University of Münster*

09/2026 **School on Exponential Motives**, *University of Duisburg–Essen*

09/2026 **Non-archimedean methods in arithmetic and tropical geometry**, *Goethe University Frankfurt*

09/2026 **New Perspectives on Stable Homotopy and Beyond**, *Max Planck Institute for Mathematics*

06/2026 **37th NRW Topology Meeting**, *University of Münster*

04/2026 **Kleine AG: Rapoport–Zink spaces and p-adic uniformization**, *University of Münster*

04/2026 **Kleine AT: Topological cyclic homology**, *University of Wuppertal*

03/2026 **Recent Advances in Motivic Cohomology**, *University of Osnabrück*

10/2025 **Kleine AT: Evenness via Spectral Sequences**, *Max Planck Institute for Mathematics*

Teaching

Courses

2026–2027 **University of Bonn**
Teaching assistant for *Algebraic Geometry I*

2025 **University of Helsinki**
Teaching assistant for *Topology IB*

2025 **University of Helsinki**
Teaching assistant for *Topology IA*

2025 **University of Helsinki**
Teaching assistant for *Linear algebra II*

2024 **University of Helsinki**
Teaching assistant for *Limits*

Other

2023–2025 **Uplus yksityisopetus**, *Helsinki*
Private mathematics teacher

Organizing

Seminars

2026–2027 **University of Bonn**
Learning seminar on moduli spaces of mixed-characteristic local shtukas (joint with Erick Miguel Díaz López)

2026–2027 **University of Bonn**
Learning seminar on topological cyclic homology (joint with Lewis Ngo)

2026 **University of Bonn**
Learning seminar on algebraic K-theory

Service

2023 **Integraldagarna**, *Helsinki*
Workshop on developing the high school mathematics curriculum in Finland

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